

# Utkarsh Persai

Phone: +91 9589573779   Email: utkarshpersai@gmail.com   LinkedIn: utkarsh-persai-2087a6282

## Summary

Computer Science undergraduate at VIT Vellore with strong foundations in software engineering, machine learning, and system design. Experienced in building scalable backend systems, real-time data pipelines, and intelligent applications integrating hardware and AI models. Proficient in Python-based development with hands-on experience in end-to-end project implementation and deployment.

## Education

**2023 – 2027   B.Tech, Computer Science and Engineering, VIT Vellore, Tamil Nadu**  
Expected graduation: 2027

## Experience

- May 2026 – Jun 2026   Industrial IoT Intern, Bharat Heavy Electricals Limited (BHEL), Bhopal**
- Completed Industrial Training at BHEL, gaining hands-on exposure to industrial automation and manufacturing operations.
  - Contributed to the MOLA Edge Gateway project, a Raspberry Pi-based industrial edge computing solution, focusing on the integration and software components of the system.
  - Worked as part of a team to connect system components and coordinate the software layer, gaining practical understanding of industrial IoT and Industry 4.0 data pipelines.
- Apr 2024 – May 2026   Core Member, IEEE CS VIT, Vellore**
- Contributed to 10+ technical and community events across development and outreach domains.
  - Led 4+ events with 100–250 attendees; managed logistics, execution, and coordination.
  - Secured 3 corporate sponsorships through structured outreach and negotiations.

## Projects

- 2026   AI-Powered Gamified Spirometry Device (1st Place IIC Hackathon Winner)**
- Designed ESP32-based portable spirometry device for real-time Pulmonary Function Testing aligned to ATS/ERS standards.
  - Built FastAPI + React platform with live WebSocket streaming and ML-based (Random Forest, Gradient Boosting) automated reporting.
  - Engineered low-cost 3D-printed solution reducing hardware cost by 98%.
  - Tech: ESP32, HX710B, FastAPI, React, Python, Scikit-learn.
- 2026   AI-Enabled Smart Agriculture System (IoT + ML)**
- Designed IoT-to-ML pipeline: ESP32 sensors -> MQTT -> Python backend -> Streamlit dashboard.
  - Implemented real-time monitoring and crop prediction model with authentication and alert system.
  - Tech: Python, ESP32, MQTT, Streamlit, scikit-learn, REST APIs.

## Certifications

**2025   AWS APAC Solutions Architecture** – Designed scalable cloud architecture using AWS services.

## Skills

|                   |                                                                                                             |
|-------------------|-------------------------------------------------------------------------------------------------------------|
| Languages & Tools | Python, SQL, C/C++, PyTorch, TensorFlow, ESP32, MQTT, Git, MySQL, Oracle DB, HTML                           |
| Domains           | Machine Learning, Deep Learning, IoT Systems, Backend Development, Explainable AI, Real-time Data Pipelines |
| Soft Skills       | Leadership, Communication, Problem Solving, Technical Documentation                                         |

## Languages Known

Hindi (Native), English (Fluent), German (Beginner)